

Tran Hieu Ngan

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EDUCATION

Ho Chi Minh University of Technology - VNU HCM

Ho Chi Minh City, Vietnam

Bachelor of Engineering subject Control Engineering and Automation

Sep 2017 – Nov 2023

- **GPA:** 3.0 / 4.0
- **Relevant Coursework:** Digital Image Processing, Digital Signal Processing, Artificial Intelligence in Control, Computer Systems and Programming Language, Fundamental of Control Systems.

WORK & EXPERIENCE

Manipal Institute of Technology

Manipal, Karnataka, India

Bioengineering Medical Image Processing Intern

July 2024 to present

- Developing software that uses **artificial intelligence** to reconstruct **4D lung images** from over **800 frames of 2D MRI data**. I measure **dynamic lung volume** by analyzing MRI slices taken during **free breathing cycles** using **MATLAB, Python, and C++**.
- Collaborating closely with the **Radiology Lab at Karatsuba Hospital in Manipal, Karnataka, India**, to collect and process MRI data for the project. My responsibilities include developing algorithms for image reconstruction and volume measurement, ensuring the data is accurate and relevant for clinical applications.
- Proposing **solutions, writing research papers, and developing research scripts** to improve lung volume estimation methods. This work aims to enhance diagnostic tools and treatment planning for patients with **Thoracic Insufficiency Syndrome (TIS)**.

Hitachi Digital Service

Ho Chi Minh City, Vietnam

Software Engineering - Full Stack Web Application Developer

Aug 2023 to July 2024

- From Aug-2023 My Team Development E-commerce web application launched in Mexico, Panama, and Chile server market of 150 million people.
- I have experience in application, GUI builder, DevOps, Unit testing, and Analysis Requirements.

Software Engineering - Python

April 2022 to Aug 2023

- Dec 2022 to Aug 2023 Guided module Image Processing for a transportation processing vehicle data ingestion and analysis Improve Safety and Performance of transportation capacity by Python and Pyspark.
- Collaborate closely with the Business Analysis to clarify requirements for the module Image Processing output and integrate the tracking system.
- At release in Aug 2023, Customers appreciated the Business outcome Teams and extended the contract to 40 members.

Ho Chi Minh University of Technology

Ho Chi Minh City, Vietnam

Project 3D Unet build 3D from 2D organs from multiple 2D Image MRI slice

July 2024 to Jan 2025

- Developed and implemented a 3D U-Net model to reconstruct 3D organ structures from multiple 2D MRI slices, enhancing medical image segmentation accuracy.
- Conducted end-to-end pipeline design, including data preprocessing, model training, and evaluation, achieving improved volumetric reconstruction and diagnostic capabilities.

Project Large Language Model

Jan to July 2024

- I deployed the LLM-Llama model with NVIDIA Jetson Xavier and Integrated Llama for a Team Chat 30 members to support answering the question about the Student Researcher Lab in Ho Chi Minh University of Technology.
- Designed a pipeline and deployed Yolov8 model object detection for self-driving cars.

Project Autonomous Car Lande Autodetection

April 2023

- Deployed segmentation on Raspberry Pi v3 to detect street and orientation for a self-driving car
- Developed an autonomous car frame and implemented a kinetic energy system-controlled microcontroller unit.

ACTIVITIES

Ho Chi Minh University of Technology Student Research Lab

Ho Chi Minh City, Vietnam

Active member

2020

- Training and workshop on basic electrical components (IC, Register, LED, PCB Design, MCU) for 200 freshers.
- Familiar with ARM Microcontroller with peripherals (UART, I2C, SPI, USB, CAN,...) as well as PCB design
- Trained basic C programming for second-year students mentored career paths in Engineering for 30 students and prepared assessments for new members.

Animal Adoption Central

2022

- Cared for animals by bathing them weekly, feeding kittens and puppies, monitoring health conditions, and ensuring timely vaccinations and veterinary check-ups.

SKILLS

Technical Skills: Python, PHP, R, C, C++, HTML, CSS, Data Analysis, Research Skills, Hardware design, Microsoft Office (Advanced Excel, Word), Git, Docker, CI/CD(basic), Pytorch, SAM 2 Machine learning model, Yolo Machine learning model,

Soft skills: Project Management, Leadership, Critical Thinking, Creative Thinking, Public Speaking, Flexibility, Multicultural Adaptability, Intercultural Communication.

Certifications & Training: [Participant Summer School for Young Leadership 2022](#), [Scale Agile Framework for Scrum master](#)

LANGUAGE

English (Toeic 610, PTE: 38), Japanese (Basic), Vietnamese (Native), Germany(Begining)